MIL-F-5373C(ASG)
24 JANUARY 1964
Superseding
MIL-F-5373B(ASG)
20 September 1956

MILITARY SPECIFICATION

FUSEHOLDERS, BLOCK TYPE, AIRCRAFT

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 This specification covers fuseholders for use in aircraft electric circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-533	Copper-Beryllium Alloy Strip
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-365	Silver Plating, Electrodeposited; General Requirements for

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-5372	Fuse, Current Limiter Type, Aircraft
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-70327	Drawings, Engineering and Associated Lists

FSC 5920

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electric Component Parts
MS20659	Terminal, Lug, Crimp Style, Copper, Uninsulated, Class 1
MS25036	Terminal, Lug, Crimp Style, Copper, Insulated, Class 1
MS25190	Wire, Electric, 600 and 3000 Volt, Copper, Aircraft, 105° C

(Copies of specifications, standards, publications, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The fuseholders furnished under this specification shall be products which have been tested and have passed the qualification tests specified herein, and have been listed on or approved for listing on the applicable Qualified Products List (see 6.3).

3.2 Data required.- Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents referenced in section 2, herein (see 6.2).

3.3 Materials.-

3.3.1 Plastic materials.- Plastic materials used on the fuseholder base shall be constructed of plastic material conforming to MIL-M-14, as indicated on the applicable military standard (MS). Plastic materials shall neither support combustion nor give off noxious gas when subjected to arcs, such as those caused by interrupting heavy short-circuit currents or explosions of gaseous vapors. Plastic materials with cellulose fillers shall not be used.

3.3.2 Protective treatment.- When materials are used in the construction of the fuseholders that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that in no way will prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

3.3.3 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143 or as specified in the contract or order, except as provided in the following paragraph.

3.3.3.1 Standard parts.- Military standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts, such as screws, bolts, nuts, and cotter pins, may be used, provided they possess suitable properties and are replaceable by the military standard parts without alteration, and provided the corresponding standard part numbers are referenced in the parts list, and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used, provided they conform to all requirements of this specification.

3.4 Design and construction.- The fuseholders shall be so designed and constructed that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Fuseholder.- The fuseholders shall be constructed in the size and rating shown on the applicable MS. The fuseholders shall be so designed that it shall not be possible to insert fuse units which are outside the rating range of the block.

3.4.2 Mounting.- The arrangement of mounting holes shall be as shown on the applicable MS. The fuseholders shall be so arranged that they can be mounted in groups and conveniently interconnected to form buses.

3.4.3 Spring clips.- Spring clips shall be of beryllium copper conforming to QQ-C-533 and shall be silver plated in accordance with type III, grade A of QQ-S-365.

3.4.4 Terminal attaching hardware.- Screws and washers shall be cadmium plated in accordance with QQ-P-416 or tin plated.

3.4.5 Life.- The fuseholders shall not sustain damage with any fuse of proper rating inserted and carrying rated current for a period of not less than 500 hours under the operating conditions as specified herein.

3.4.6 Spacing and creepage distances.- The minimum clearance in air (phase-to-phase and phase-to-ground) shall be 1/8 inch. The minimum creepage distance on insulation surfaces shall be 1/4 inch.

3.4.7 Terminals.- The fuseholders shall accommodate terminal lugs conforming to MS20659 or MS25036 of the proper size as determined by the fuseholder rating (see 4.5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

3.4.8 Random drop.- The fuseholder, with a limiter in accordance with MIL-F-5372 inserted, shall sustain no physical damage, and the limiter shall remain in proper position with respect to the spring clips when subjected to the random drop test specified in 4.7.5.

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-70327.

3.6 Weight and dimensions.- The weight and dimensions shall be as shown on the applicable MS.

3.7 Screw threads.- Screw threads shall be in accordance with MIL-S-7742.

3.8 Performance.- The fuseholders shall be capable of operating under the following conditions (see 4.7.3 and 4.7.4):

(a) Temperature:

(1) Low temperature: A minimum ambient temperature of -65° C from sea level to 60,000 feet altitude.
(2) High temperature: A maximum ambient temperature of 125° C at sea level, varying uniformly to 2° C at 60,000 feet altitude.

(b) Altitude: A pressure altitude range of from sea level to 60,000 feet.
(c) Humidity: A relative humidity of from 0.5 to 100 percent.
(d) Sand and dust: Exposure to airborne sand particles encountered in service.
(e) Vibration: Conditions of vibration incident to normal continuous use in aircraft.
(f) Salt spray: Exposure to salt-laden atmosphere.

3.9 Identification of product.- The following information shall be plainly marked on each fuseholder:

Ampere rating
MS Part No.
Manufacturer's name or trademark

3.9.1 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

3.10 Workmanship.- The fuseholders, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness in marking of parts and assemblies and freedom of parts from burrs and sharp edges.

3.10.1 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices.

3.10.2 Cleaning.- The fuseholders shall be thoroughly cleaned of loose metal chips and other foreign material after final assembly. Burrs and sharp edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of fuseholders shall be classified as follows:

(a) Qualification tests (4.3)
(b) Quality conformance tests (4.4)

4.3 Qualification tests.-

4.3.1 Sampling instructions.- Qualification test samples shall consist of six fuseholders of each rating. Samples shall be identified as required and forwarded to the activity responsible for qualification designated in the letter of authorization from that activity (see 6.3).

4.3.2 Tests.- The qualification tests shall consist of all the tests specified in 4.6 and 4.7.

4.4 Quality conformance tests.- Quality conformance tests shall consist of individual tests (4.4.1) and sampling tests (4.4.2).

4.4.1 Individual tests.- Each fuseholder submitted for acceptance under contract shall be subjected to the following test specified in 4.6. In addition, each fuseholder shall be subjected to any of the tests specified herein which are deemed necessary to determine compliance with the requirements of this specification.

Examination of product (4.6.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

4.4.2 Sampling tests.- Two fuseholders of each rating shall be selected at random from each lot of 400 or fraction thereof and subjected to the insulation test specified in 4.7.1. A lot shall consist of fuseholders manufactured under essentially the same conditions.

4.4.3 Rejection.- When any item from a lot fails the specified tests, this shall be cause for rejection of the lot.

4.5 Test conditions.-

4.5.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 30 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.5.2 Test leads.- All tests which are made with fuseholders containing fuses and with fuses carrying current shall be accomplished with one conductor on each terminal. The size of conductors for test purposes shall be as follows and shall conform to MS25190:

Fuse rating (amp)	Cable size
5	MS25190-B-20
10	MS25190-B-18
20	MS25190-B-16
30	MS25190-B-14
40	MS25190-B-12
50	MS25190-B-10
60	MS25190-B-8
80	MS25190-B-6
100	MS25190-B-4
150	MS25190-B-2
200	MS25190-B-1
300	MS25190-B-3-0
400	MS25190-B-4-0
500	MS25190-B-2(2-0)

4.6 Examinations.-

4.6.1 Examination of product.- The fuseholder shall be inspected to determine compliance with this specification with respect to materials, workmanship, construction, and dimensions.

4.6.2 Packaging, packing, and marking.- Preparation for delivery shall be examined for conformance with section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

4.7 Test methods.-

4.7.1 Insulation.- The fuseholder assembled to a grounded metal plate and containing fuses shall be subjected to an overpotential test as follows:

A potential of 2,500 volts (V) root mean square (rms), alternating current (ac), at commercial frequency shall be applied for 1 minute between each phase of the fuseholder and between each phase and ground. There shall be no breakdown of insulation of either fuses or fuseholder as a result of this test.

4.7.2 Temperature rise.- The fuseholder containing fuses of the maximum rating specified for the holder shall be subjected to the following duty cycle at 25° C and shall be repeated 25 times:

Percent load	Time in minutes
100	30
140	1
0	29

The maximum temperature rise at any given point on the fuseholder during the 25th cycle shall not differ by more than 10 percent from the maximum temperature rise experienced at the same point during the first cycle, the ambient temperature being maintained at 25° C throughout the test.

4.7.3 Environmental.- Three fuseholders of each rating shall be subjected to each of the following tests. There shall be no cracking, loosening of parts, corrosion of metal parts, or other visible damage as a result of these tests:

(a) Temperature cycling: Method 102A, condition D of MIL-STD-202.

(b) Humidity: Procedure I of MIL-E-5272. Immediately after completion of the humidity test, a potential of 1,000V rms, ac, at commercial frequency, shall be applied for 1 minute between phase and ground and between phases of the fuseholder when the fuseholder is attached to a grounded metal plate with typical mounting screws. There shall be no breakdown of insulation as a result of this test.

(c) Salt spray: Method 101B, condition B of MIL-STD-202.

(d) Sand and dust: Procedure I, Part I of MIL-E-5272.

4.7.4 Vibration.- The fuseholder containing limiters shall be subjected to the vibration test Procedure XII of MIL-E-5272 for a period of 6 hours in each of the two positions most likely to result in loosening of parts. The test shall be made with the leads of the largest size specified for the fuseholder connected to the holder and anchored outside of the test platform 6 inches from the terminals of the holder. The fuseholder shall withstand this test without damage and without loosening of either the terminals or the fuse blades within the spring clips.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

4.7.5 Random drop.- A limiter of proper size and in accordance with MIL-F-5372 shall be inserted in the limiter holder under test. The assembly shall be rigidly mounted in a mounting fixture and subjected to the random drop test in accordance with Method 203A of MIL-STD-202.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.- Fuseholders shall be preserved, packaged, and packed in accordance with the applicable methods and levels of MIL-E-17555, as specified in the contract or order (see 6.2). Marking of interior packages and exterior shipping containers shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use.- The fuseholders covered by this specification are intended for use in holding fuses conforming to MIL-F-5372 in aircraft electrical circuits.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) MS Part No. of the fuseholder desired.
(c) Data requirements (see 3.2).
(d) Applicable method of preservation and unit packaging and the level of packing required (see 5.1).
(e) Intermediate packaging requirements (see 5.1).

6.3 Qualification.- With respect to products requiring qualification (see 4.3.1), awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Navy Department, Washington, D. C., 20360; however, information pertaining to qualification may be obtained from the Commanding Officer, U. S. Naval Air Development Center, Johnsville, Pennsylvania.

6.4 Marginal indicia.- The margins of this specification are marked to indicate where changes, deletions, or additions to the previous issue have been made. This is done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

Custodians:
Navy - Weps
Air Force - ASD (11)

Preparing activity:
Air Force - ASD (11)

Reviewer activity:
Navy - Weps
Air Force - ASD (11)

User activity:
Navy - Weps
Air Force - ASD (11)

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DoD Index of Specifications and Standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-F-5373C(ASG) Fuseholders, Block Type, Aircraft

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS OF ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Aeronautical Standards Group (ASG)

Stop 48

Washington, D. C. 20390

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:32Z
Check the source to verify that this is the current version before use.